Lesson 1: Mixed and Entire Radicals

Assignment

1. Without a calculator, evaluate where possible.

(a) $\sqrt{144}$ (b) $-\sqrt{81}$ (c) $\sqrt{-36}$

(d) $\sqrt[3]{-343}$ (e) $\sqrt[4]{625}$ (f) $\sqrt[4]{-1}$

(g) $\sqrt[5]{-100\,000}$ (h) $\sqrt[3]{\frac{64}{125}}$ (i) $\sqrt{1.44}$

2. State whether each statement is true or false. Correct each false statement.

(a) The square roots of 49 are 7 and -7. (b) $\sqrt{49} = \pm 7$

(c) -27 has no cube root. (d) $\sqrt[4]{-16}$ is not a real number.

(e) If $x^2 = 25$, then $x = 5$ or $x = -5$. (f) $\sqrt[3]{-8} = -\sqrt[3]{8}$

3. (a) Classify each number as rational or irrational.

(i) $\sqrt{45}$ (ii) $\sqrt[3]{-64}$ (iii) $\sqrt{\frac{25}{36}}$ (iv) $\sqrt[4]{32}$ (v) $\sqrt{0.9}$ (vi) $\sqrt[3]{0.008}$

(b) Between which two consecutive whole numbers does each lie? $\sqrt{90}$, $\sqrt[3]{90}$, $\sqrt[4]{90}$

(c) Use (b) to arrange the three numbers from least to greatest.

4. State whether each statement is true or false. Show a check.

(a) $\sqrt{5} \times \sqrt{7} = \sqrt{35}$ (b) $\sqrt{16+9} = \sqrt{16} + \sqrt{9}$ (c) $\dfrac{\sqrt{48}}{\sqrt{3}} = 4$

(d) $\sqrt{100} - \sqrt{36} = \sqrt{64}$ (e) $\sqrt[3]{9} \times \sqrt[3]{3} = 3$ (f) $\sqrt{4 \times 25} = \sqrt{4} \times \sqrt{25}$

5. Write each radical as a mixed radical in simplest form.

(a) $\sqrt{28}$ (b) $\sqrt{63}$ (c) $\sqrt{150}$ (d) $\sqrt{242}$

(e) $4\sqrt{45}$ (f) $\sqrt[3]{56}$ (g) $\sqrt[3]{-135}$ (h) $\sqrt[3]{192}$

(i) $\sqrt[4]{48}$ (j) $-3\sqrt[3]{81}$ (k) $\sqrt[4]{1250}$ (l) $-\sqrt{500}$

6. Simplify. Use prime factors if it helps.

(a) $\sqrt{3200}$ (b) $\sqrt{1176}$ (c) $\sqrt[3]{4000}$ (d) $\sqrt[3]{-2662}$

7. Each student's answer is not correct or not in simplest form. Explain the mistake and give the correct simplest form.

(a) Jade: $\sqrt{180} = 3\sqrt{20}$

(b) Liam: $\sqrt[3]{48} = 4\sqrt[3]{3}$

8. Given $\sqrt{7} \approx 2.646$, determine an approximate value of each without a calculator.

(a) $\sqrt{700}$ (b) $\sqrt{28}$ (c) $\sqrt{0.07}$ (d) $\sqrt{63}$ (e) $\sqrt{70\,000}$

9. Write each mixed radical as an entire radical.

(a) $6\sqrt{2}$ (b) $3\sqrt{11}$ (c) $-4\sqrt{5}$ (d) $2\sqrt[3]{7}$ (e) $-3\sqrt[3]{4}$

(f) $10\sqrt[3]{3}$ (g) $2\sqrt[4]{3}$ (h) $\frac{1}{3}\sqrt{54}$ (i) $\frac{1}{2}\sqrt[3]{40}$

10. Without a calculator, arrange from greatest to least: $4\sqrt{5},\ 6\sqrt{2},\ 3\sqrt{7},\ 2\sqrt{19},\ \sqrt{77}$

11. Consider the radicals $2\sqrt[3]{11},\ 3\sqrt[3]{3},\ \sqrt[3]{85},\ -4\sqrt[3]{2},\ -2\sqrt[3]{15}$.
(a) Explain how to order them without a calculator.

(b) Arrange them from least to greatest.

12. Give each answer exactly, in simplest form.
(a) A square tile has an area of 180 cm^2. Determine its side length and its perimeter.

(b) A square has twice the area of a square with side 10 m. Determine its side length.

(c) A cube has a volume of 448 cm^3. Determine its edge length.

13. Determine each length exactly, in simplest form.
(a) The hypotenuse of a right triangle with legs 6 cm and 9 cm.

(b) The distance between $P(1, -6)$ and $Q(-4, 4)$.

14. Which radical is in simplest form?
A. $\sqrt{40}$ **B.** $\sqrt{42}$ **C.** $\sqrt[3]{24}$ **D.** $\sqrt[4]{32}$

15. As an entire radical, $-2\sqrt[3]{5}$ is
A. $\sqrt[3]{-40}$ **B.** $-\sqrt[3]{10}$ **C.** $\sqrt[3]{40}$ **D.** $\sqrt[3]{-10}$

16. Which statement is true?

A. $\sqrt{9} + \sqrt{16} = \sqrt{25}$
B. $\sqrt{2} \times \sqrt{8} = 4$
C. $\sqrt{50} - \sqrt{25} = \sqrt{25}$
D. $\sqrt[3]{8} + \sqrt[3]{8} = \sqrt[3]{16}$

17. Numerical Response A cube has a volume of 1372 cm^3. Its edge length in simplest form is $p\sqrt[3]{q}$ cm. The value of $p + q$ is _____.

18. Numerical Response When $5\sqrt{6}$ is written as the entire radical $\sqrt{n}$, the value of n is _____.

Lesson 2: Radicals with Variables and Their Restrictions

Assignment

1. State the restriction on the variable.

(a) $\sqrt{x+5}$ (b) $\sqrt{x-8}$ (c) $\sqrt{3x-12}$

(d) $\sqrt{10-5x}$ (e) $\sqrt{-4x}$ (f) $\sqrt[3]{2x+9}$

(g) $\sqrt[4]{x-1}$ (h) $\sqrt[5]{7-x}$ (i) $\sqrt{\frac{1}{2}x+3}$

2. For what values of x is each expression defined? Explain briefly.

(a) $\sqrt{x^2+9}$ (b) $\sqrt{-x^3}$ (c) $\dfrac{6}{\sqrt{x-1}}$ (d) $\sqrt[3]{x^2-5}$

3. Consider $\sqrt{2x+10}$.

(a) State the restriction on x.

(b) Evaluate exactly, in simplest form, where possible: $x=-3$, $x=7$, $x=20$, $x=-6$.

4. (a) Evaluate $\sqrt{x^2}$ for $x=6$ and for $x=-6$. (b) Explain why we write "$\sqrt{x^2}=x,\ x\geq 0$".

5. Simplify. Assume variables under an even index are ≥ 0.

(a) $\sqrt{x^{10}}$ (b) $\sqrt{a^{11}}$ (c) $\sqrt[3]{y^{12}}$

(d) $\sqrt[3]{m^8}$ (e) $\sqrt[4]{x^{13}}$ (f) $\sqrt{x^3y^6}$

6. Write each radical in simplest form. State the restrictions where needed.

(a) $\sqrt{20x^2}$ (b) $\sqrt{27a^3}$ (c) $\sqrt{98m^5n^2}$

(d) $\sqrt{150x^7y^4}$ (e) $\sqrt[3]{16a^5}$ (f) $\sqrt[3]{-250x^6y^8}$

(g) $\sqrt[3]{81m^{10}}$ (h) $\sqrt[4]{162x^5}$ (i) $-\sqrt{63a^9b}$

7. Each simplification has an error. Explain it and give the correct answer ($x, a \geq 0$).

(a) $\sqrt{36x^{16}} = 6x^4$ (b) $\sqrt[3]{27a^6} = 9a^2$ (c) $\sqrt{9+x^2} = 3+x$

8. Write each mixed radical as an entire radical. Assume variables under an even index are ≥ 0.

(a) $5x\sqrt{x}$ (b) $2a^2\sqrt{3a}$ (c) $-y\sqrt{7y}$

(d) $3m\sqrt[3]{m^2}$ (e) $-2a^2\sqrt[3]{5a}$ (f) $xy^2\sqrt{2x}$

9. A square has an area of $75a^4b^2$ square units ($a, b > 0$). Determine its side length and perimeter in simplest form.

10. A cube has a volume of $128x^7$ cubic units. Determine its edge length in simplest form.

11. The legs of a right triangle are $2x$ and $4x$, $x > 0$. Determine the length of the hypotenuse in simplest form.

12. The restriction on the variable in $\sqrt{6-3x}$ is

A. $x \geq 2$ **B.** $x \leq 2$ **C.** $x \geq -2$ **D.** $x \leq -2$

13. In simplest form, $\sqrt{48a^7}$, $a \geq 0$, is

A. $4a^3\sqrt{3a}$ **B.** $16a^3\sqrt{3a}$ **C.** $2a^3\sqrt{12a}$ **D.** $4a^6\sqrt{3a}$

14. Which radical is defined for every real number x?

A. $\sqrt{x-1}$ **B.** $\sqrt[3]{x-1}$ **C.** $\sqrt[4]{x}$ **D.** $\sqrt{-x}$

15. Numerical Response In simplest form, $\sqrt{200x^6y^9} = ax^by^c\sqrt{2y}$ $(x, y \geq 0)$. The value of $a+b+c$ is _____.

16. Numerical Response The largest integer value of x for which $\sqrt{50-4x}$ is defined is _____.

Lesson 3: Adding and Subtracting Radicals

Assignment

1. Simplify. State the restriction in (f).
(a) $8\sqrt{6} - 3\sqrt{6}$ (b) $2\sqrt{15} + 7\sqrt{15} - 12\sqrt{15}$ (c) $4\sqrt[3]{5} + \sqrt[3]{5} - 9\sqrt[3]{5}$

(d) $6\sqrt{2} - 4\sqrt{7} + \sqrt{2} + 9\sqrt{7}$ (e) $10 - 3\sqrt{3} + 4 + 7\sqrt{3}$ (f) $5\sqrt{y} + 8\sqrt{y} - \sqrt{y}$

2. (a) A student wrote $\sqrt{12} + \sqrt{3} = \sqrt{15}$. Explain the error and give the correct answer in simplest form.

(b) Which pairs are like radicals once they are in simplest form? Explain each.
(i) $\sqrt{12}$ and $\sqrt{27}$ (ii) $3\sqrt{5}$ and $5\sqrt{3}$ (iii) $\sqrt{2}$ and $\sqrt[3]{2}$ (iv) $\sqrt{18}$ and $\sqrt{50}$ (v) $\sqrt[3]{16}$ and $\sqrt[3]{54}$

3. Write each expression as a single radical in simplest form.
(a) $\sqrt{28} + \sqrt{63}$ (b) $\sqrt{200} - \sqrt{72}$ (c) $3\sqrt{20} - \sqrt{125}$

(d) $\sqrt{44} + 2\sqrt{99} - \sqrt{176}$ (e) $4\sqrt{54} - 2\sqrt{24} - \sqrt{150}$ (f) $-2\sqrt{243} + 5\sqrt{48}$

4. Simplify.
(a) $\sqrt{45} + \sqrt{8} - \sqrt{20} + \sqrt{50}$ (b) $\sqrt{112} - \sqrt{18} + \sqrt{63} - \sqrt{32}$

(c) $3\sqrt{24} + \sqrt{90} - \sqrt{150}$ (d) $2\sqrt{96} - \sqrt{300} + 4\sqrt{27}$

5. Simplify.

(a) $\sqrt[3]{24} + \sqrt[3]{192}$

(b) $5\sqrt[3]{16} - \sqrt[3]{250}$

(c) $\sqrt[3]{-128} + \sqrt[3]{54} + \sqrt[3]{686}$

(d) $\sqrt[4]{80} + \sqrt[4]{405}$

(e) $\sqrt[3]{40} + \sqrt{40} - \sqrt[3]{5}$

(f) $2\sqrt[4]{32} - \sqrt[4]{162}$

6. Simplify.

(a) $\frac{1}{2}\sqrt{20} + \frac{1}{3}\sqrt{45}$

(b) $\frac{2}{5}\sqrt{75} - \frac{1}{4}\sqrt{48}$

(c) $\frac{3}{4}\sqrt{8} + \frac{1}{6}\sqrt{72}$

(d) $\frac{1}{2}\sqrt[3]{16} + \frac{2}{3}\sqrt[3]{54}$

7. Simplify. State the restrictions on the variables.

(a) $\sqrt{16x} - \sqrt{x} + \sqrt{49x}$

(b) $\sqrt{8a^3} + a\sqrt{18a}$

(c) $\sqrt[3]{24m^4} + m\sqrt[3]{81m}$

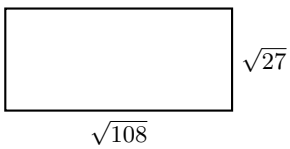
(d) $\sqrt{75y} - \sqrt{12y} + \sqrt{20y}$

(e) $\sqrt{9x - 18} + 4\sqrt{x - 2}$

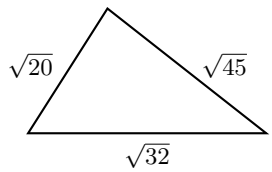
(f) $\sqrt{50xy^2} - y\sqrt{8x}$

8. Determine the perimeter of each figure in simplest form.

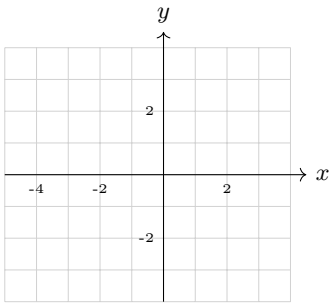
(a) Rectangle



(b) Triangle



9. Plot $P(-4, 1)$, $Q(2, 3)$ and $R(0, -3)$ and join them. Determine the perimeter of $\triangle PQR$ in simplest form.



10. (a) Point C lies on segment AB. If $AB = \sqrt{300}$ and $AC = \sqrt{48}$, determine CB in simplest form.

(b) A rectangle has perimeter $16\sqrt{5} + 6\sqrt{3}$ m and length $5\sqrt{5} + \sqrt{12}$ m. Determine its width in simplest form.

11. If $2x - \sqrt{12} = \sqrt{108}$, determine x as a mixed radical in simplest form and as an entire radical.

12. Each sequence is arithmetic.
(a) $\sqrt{2},\ \sqrt{8},\ \sqrt{18}, \ldots$ Determine d and the next two terms in simplest form.

(b) $10 - \sqrt{3},\ 8 + \sqrt{3},\ 6 + 3\sqrt{3}, \ldots$ Determine d and t_{20}.

13. Which radical can be combined with $\sqrt{20}$ by addition?
A. $\sqrt{40}$ **B.** $\sqrt{45}$ **C.** $\sqrt{50}$ **D.** $2\sqrt[3]{5}$

14. In simplest form, $\sqrt{50} + \sqrt{18} - \sqrt{8}$ is
A. $6\sqrt{2}$ **B.** $\sqrt{60}$ **C.** $10\sqrt{2}$ **D.** $2\sqrt{15}$

15. A square has sides of length $\sqrt{12} + \sqrt{3}$ cm. Its perimeter is
A. $3\sqrt{3}$ cm **B.** $4\sqrt{15}$ cm **C.** $12\sqrt{3}$ cm **D.** $6\sqrt{3}$ cm

16. Numerical Response A square has sides of length $\sqrt{98} + \sqrt{8}$ m. Its perimeter can be written as $a\sqrt{2}$ m. The value of a is _____.

17. Numerical Response When $\frac{1}{2}\sqrt[3]{54} + \sqrt[3]{16}$ is written in the form $a\sqrt[3]{2}$, the value of a, as a decimal, is _____.

Lesson 4: Multiplying Radicals

Assignment

1. Multiply and simplify.

(a) $\sqrt{5} \cdot \sqrt{7}$ (b) $(2\sqrt{3})(6\sqrt{11})$ (c) $(-4\sqrt{2})(3\sqrt{14})$

(d) $\sqrt{8} \cdot \sqrt{18}$ (e) $(5\sqrt{6})(2\sqrt{6})$ (f) $(3\sqrt{10})(-\sqrt{15})$

(g) $(2\sqrt{21})(\sqrt{35})$ (h) $\sqrt{20} \cdot \sqrt{45}$ (i) $(-3\sqrt{7})(-2\sqrt{7})$

(j) $\left(\sqrt[3]{9}\right)\left(\sqrt[3]{3}\right)$ (k) $\left(4\sqrt[3]{2}\right)\left(-\sqrt[3]{20}\right)$ (l) $\left(2\sqrt[4]{8}\right)\left(\sqrt[4]{6}\right)$

2. Simplify.

(a) $\left(\sqrt{11}\right)^2$ (b) $\left(3\sqrt{2}\right)^2$ (c) $-\left(4\sqrt{5}\right)^2$ (d) $\left(-2\sqrt{3}\right)^2$ (e) $\left(\sqrt[3]{4}\right)^3$ (f) $\left(2\sqrt[3]{5}\right)^3$

3. Multiply and simplify.

(a) $\left(\sqrt{2}\right)\left(\sqrt{3}\right)\left(\sqrt{6}\right)$ (b) $\left(2\sqrt{5}\right)\left(-\sqrt{10}\right)\left(3\sqrt{2}\right)$

(c) $\left(\sqrt{3}\right)\left(2\sqrt{6}\right)\left(4\sqrt{8}\right)$ (d) $\left(\sqrt[3]{4}\right)\left(\sqrt[3]{6}\right)\left(\sqrt[3]{9}\right)$

4. Write each as a product of two mixed radicals in two different ways. (a) $10\sqrt{6}$ (b) $24\sqrt{15}$

5. Multiply and simplify. State the restrictions on the variables.

(a) $(4\sqrt{a})\left(2\sqrt{b}\right)$ (b) $\left(\sqrt{3x}\right)\left(\sqrt{12x}\right)$ (c) $\left(2\sqrt{5y^3}\right)\left(3\sqrt{10y}\right)$

(d) $\left(\sqrt[3]{2n^2}\right)\left(\sqrt[3]{4n^2}\right)$ (e) $\left(\sqrt{6x}\right)\left(\sqrt{2x^2}\right)$ (f) $\left(\sqrt{x+3}\right)^2$

6. Expand and simplify. State the restrictions in (e).

(a) $\sqrt{2}\left(\sqrt{6}-5\right)$ (b) $3\sqrt{5}\left(2\sqrt{5}+\sqrt{3}\right)$

(c) $-\sqrt{6}\left(\sqrt{3}-2\sqrt{2}\right)$ (d) $2\sqrt{3}\left(\sqrt{12}-4\sqrt{6}+\sqrt{27}\right)$

(e) $\sqrt{x}\left(2\sqrt{x}-\sqrt{y}\right)$ (f) $\sqrt[3]{2}\left(\sqrt[3]{4}+3\sqrt[3]{12}\right)$

7. Expand and simplify. State the restriction in (e).

(a) $\left(2+\sqrt{3}\right)\left(4+\sqrt{3}\right)$ (b) $\left(5-\sqrt{2}\right)\left(1+3\sqrt{2}\right)$

(c) $\left(\sqrt{6}+2\sqrt{5}\right)\left(3\sqrt{6}-\sqrt{5}\right)$ (d) $\left(3\sqrt{2}-\sqrt{6}\right)\left(\sqrt{3}+2\sqrt{2}\right)$

(e) $\left(\sqrt{x}+4\right)\left(\sqrt{x}-1\right)$

8. Expand and simplify. State the restrictions in (e).

(a) $\left(\sqrt{7}-2\right)^2$ (b) $\left(3+2\sqrt{5}\right)^2$ (c) $\left(4\sqrt{2}-\sqrt{6}\right)^2$

(d) $2\left(\sqrt{10}+\sqrt{5}\right)^2$ (e) $\left(2\sqrt{a}+\sqrt{b}\right)^2$

9. Write the conjugate of each expression, then multiply the expression by its conjugate. State the restriction in (f).

(a) $\sqrt{11}-3$ (b) $5+\sqrt{2}$ (c) $2\sqrt{7}+3\sqrt{2}$

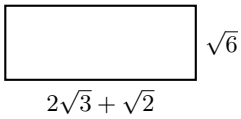
(d) $\sqrt{15}-\sqrt{6}$ (e) $3\sqrt{2}-2\sqrt{5}$ (f) $2\sqrt{x}+5$

10. (a) Expand $\left(\sqrt{6}+\sqrt{2}\right)\left(\sqrt{6}-\sqrt{2}\right)$ and $\left(\sqrt{6}+\sqrt{2}\right)^2$. Which product is rational? Explain why, using the middle terms.

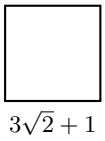
(b) Give two irrational numbers whose product is rational, and two irrational numbers whose product is irrational.

11. Determine the area of each figure in simplest form. Dimensions are in metres.

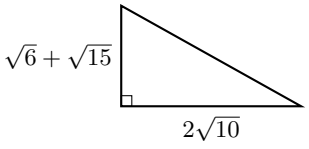
(a) Rectangle



(b) Square



(c) Triangle



12. A rectangle measures $\left(\sqrt{3}+1\right)$ cm by $\left(\sqrt{3}-1\right)$ cm.

(a) Determine its area.

(b) Determine the exact length of its diagonal in simplest form.

13. Simplify $\left(\sqrt{3}+2\right)^2-\left(\sqrt{3}-2\right)^2$.

14. For $a \geq 0$ and $b \geq 0$, the expression $\left(\sqrt{a}+\sqrt{b}\right)^2$ is equal to

A. $a+b$ **B.** $a+b+2\sqrt{ab}$ **C.** $a+b+\sqrt{ab}$ **D.** a^2+b^2

15. In simplest form, $\left(2\sqrt{6}\right)\left(3\sqrt{15}\right)$ is

A. $6\sqrt{21}$ **B.** $18\sqrt{10}$ **C.** $5\sqrt{90}$ **D.** $3\sqrt{10}$

16. Which product is a rational number?

A. $\left(\sqrt{5}+1\right)\left(\sqrt{5}+1\right)$

B. $\left(2\sqrt{3}-\sqrt{2}\right)\left(2\sqrt{3}+\sqrt{2}\right)$

C. $\left(\sqrt{7}\right)\left(\sqrt{14}\right)$

D. $\left(3+\sqrt{2}\right)\left(3\sqrt{2}\right)$

17. Numerical Response The product $\left(\sqrt{3}+2\right)\left(4\sqrt{3}-1\right)$ can be written in the form $a+b\sqrt{3}$. The value of $a+b$ is _____.

18. Numerical Response A square has sides of length $\left(3\sqrt{2}+\sqrt{6}\right)$ cm. Its area can be written as $a+b\sqrt{3}$ cm^2. The value of $a+b$ is _____.

Lesson 5: Dividing Radicals and Rationalizing Monomial Denominators

Assignment

1. Divide and write in simplest form.

(a) $\dfrac{\sqrt{72}}{\sqrt{2}}$ (b) $\dfrac{\sqrt{60}}{\sqrt{5}}$ (c) $\dfrac{15\sqrt{42}}{5\sqrt{7}}$

(d) $\dfrac{-24\sqrt{30}}{8\sqrt{3}}$ (e) $\dfrac{\sqrt[3]{192}}{\sqrt[3]{3}}$ (f) $\dfrac{12\sqrt[3]{80}}{4\sqrt[3]{2}}$

2. Simplify. (The radicands do not divide evenly.)

(a) $\dfrac{\sqrt{75}}{\sqrt{27}}$ (b) $\dfrac{3\sqrt{128}}{4\sqrt{50}}$ (c) $\dfrac{10\sqrt{63}}{6\sqrt{28}}$

(d) $\dfrac{2\sqrt[3]{250}}{\sqrt[3]{16}}$ (e) $\dfrac{5\sqrt{108}}{2\sqrt{12}}$

3. Divide each term of the numerator by the denominator, and simplify.

(a) $\dfrac{\sqrt{66}+\sqrt{30}}{\sqrt{6}}$ (b) $\dfrac{15\sqrt{14}-10\sqrt{35}}{5\sqrt{7}}$

(c) $\dfrac{4\sqrt{90}+6\sqrt{50}}{2\sqrt{5}}$ (d) $\dfrac{9\sqrt{22}+6\sqrt{55}-3\sqrt{11}}{3\sqrt{11}}$

4. Rationalize the denominator and simplify.

(a) $\dfrac{1}{\sqrt{5}}$ (b) $\dfrac{8}{\sqrt{2}}$ (c) $\dfrac{\sqrt{3}}{\sqrt{11}}$

(d) $\dfrac{15}{2\sqrt{3}}$ (e) $\dfrac{-4}{3\sqrt{6}}$ (f) $\dfrac{18}{\sqrt{12}}$

(g) $\dfrac{2\sqrt{5}}{\sqrt{8}}$ (h) $\dfrac{7}{\sqrt{98}}$ (i) $\dfrac{6\sqrt{2}}{5\sqrt{6}}$

5. Write each expression in simplest form (no fraction under a radical).

(a) $\sqrt{\dfrac{2}{3}}$ (b) $\sqrt{\dfrac{9}{7}}$ (c) $\sqrt{\dfrac{5}{12}}$

(d) $4\sqrt{\dfrac{3}{8}}$ (e) $6\sqrt{\dfrac{7}{18}}$ (f) $-\sqrt{\dfrac{45}{20}}$

6. Rationalize the denominator and simplify.

(a) $\dfrac{5+\sqrt{3}}{\sqrt{3}}$ (b) $\dfrac{\sqrt{10}-4}{\sqrt{2}}$

(c) $\dfrac{6\sqrt{2}+3\sqrt{6}}{3\sqrt{3}}$ (d) $\dfrac{\sqrt{15}-2\sqrt{6}}{2\sqrt{5}}$

7. Consider $\dfrac{9\sqrt{10}+6\sqrt{14}}{3\sqrt{2}}$.

(a) Simplify it by dividing each term of the numerator by the denominator.

(b) Simplify it by rationalizing the denominator first. Do you get the same answer?

(c) Explain why dividing each term would not work neatly for $\dfrac{9\sqrt{10}+6\sqrt{14}}{3\sqrt{5}}$, then simplify that expression.

8. Simplify and write in lowest terms.

(a) $\dfrac{6\sqrt{12}-4\sqrt{18}}{\sqrt{6}}$ (b) $\dfrac{14\sqrt{27}-6\sqrt{75}}{4\sqrt{12}}$ (c) $\dfrac{12-\sqrt{48}}{2\sqrt{3}}$

9. Rationalize the denominator and simplify.

(a) $\dfrac{2}{\sqrt[3]{5}}$ (b) $\dfrac{12}{\sqrt[3]{4}}$ (c) $\dfrac{5}{\sqrt[3]{9}}$

(d) $\sqrt[3]{\dfrac{2}{3}}$ (e) $\dfrac{8}{\sqrt[3]{-16}}$ (f) $\dfrac{10}{\sqrt[4]{8}}$

10. Simplify. State the restrictions on the variables.

(a) $\dfrac{\sqrt{63x^3}}{\sqrt{7x}}$ (b) $\dfrac{\sqrt{10x^3}+\sqrt{15x}}{\sqrt{5x}}$ (c) $\dfrac{4}{\sqrt{2x}}$

(d) $\dfrac{5y}{\sqrt{10y}}$ (e) $\sqrt{\dfrac{3}{2m}}$ (f) $\dfrac{6}{\sqrt[3]{3x}}$

11. A triangular sail has an area of $(4\sqrt{30}+6\sqrt{10})$ m^2 and a base of $2\sqrt{5}$ m. Use $A=\frac{1}{2}bh$ to determine the exact height of the sail.

12. A rectangular box has a volume of $30\sqrt{2}$ cm^3, a length of $\sqrt{6}$ cm and a width of $\sqrt{15}$ cm. Determine its exact height.

13. The height of an equilateral triangle is $\dfrac{\sqrt{3}}{2}$ times its side length. An equilateral triangle has a height of 6 cm. Determine the exact side length and perimeter.

14. Solve each equation. Write x in simplest form.
(a) $x\sqrt{5}=10$ (b) $x\sqrt{3}-\sqrt{12}=\sqrt{75}$ (c) $2\sqrt{3}\,x-\sqrt{2}=0$

15. Which expression is **not** equal to $\dfrac{\sqrt{6}}{2}$?

A. $\sqrt{\dfrac{3}{2}}$ **B.** $\dfrac{3}{\sqrt{6}}$ **C.** $\dfrac{2\sqrt{3}}{\sqrt{8}}$ **D.** $\dfrac{\sqrt{3}}{2\sqrt{2}}$

16. When $\dfrac{6}{\sqrt[3]{3}}$ is written with a rational denominator, the result is

A. $2\sqrt[3]{3}$ **B.** $2\sqrt[3]{9}$ **C.** $6\sqrt[3]{9}$ **D.** $3\sqrt[3]{3}$

17. For $x > 0$, $\dfrac{\sqrt{20x^3}}{\sqrt{5x}}$ simplifies to

A. $2x$ **B.** $4x$ **C.** $2x^2$ **D.** $2x\sqrt{x}$

18. Numerical Response The expression $\dfrac{12}{\sqrt{6}} - \dfrac{\sqrt{3}}{\sqrt{2}}$ can be written as $k\sqrt{6}$. The value of k, as a decimal, is _____.

19. Numerical Response If $\dfrac{24}{\sqrt{18}} - \dfrac{\sqrt{6}}{\sqrt{3}} = k\sqrt{2}$, then the value of k is _____.

Lesson 6: Rationalizing Binomial Denominators with Conjugates

Assignment

1. Write the conjugate of each binomial, and find the product of the binomial and its conjugate.

(a) $5 + \sqrt{2}$ (b) $\sqrt{11} - 3$ (c) $\sqrt{10} + \sqrt{3}$

(d) $4\sqrt{2} - 5$ (e) $2\sqrt{7} + 3\sqrt{2}$ (f) $-1 - \sqrt{6}$

2. Rationalize the denominator and simplify.

(a) $\dfrac{1}{\sqrt{3}+1}$ (b) $\dfrac{4}{\sqrt{5}-1}$ (c) $\dfrac{7}{3+\sqrt{2}}$

(d) $\dfrac{2}{\sqrt{6}-\sqrt{2}}$ (e) $\dfrac{12}{\sqrt{7}+\sqrt{3}}$ (f) $\dfrac{-6}{4-\sqrt{10}}$

3. Rationalize the denominator and simplify.

(a) $\dfrac{\sqrt{2}}{\sqrt{2}+1}$ (b) $\dfrac{\sqrt{5}}{\sqrt{5}-\sqrt{3}}$

(c) $\dfrac{3\sqrt{2}}{2\sqrt{3}-\sqrt{6}}$ (d) $\dfrac{2\sqrt{5}}{\sqrt{10}+2\sqrt{5}}$

4. Rationalize the denominator and simplify.

(a) $\dfrac{\sqrt{7}+2}{\sqrt{7}-2}$ (b) $\dfrac{3-\sqrt{2}}{3+\sqrt{2}}$ (c) $\dfrac{\sqrt{5}+\sqrt{2}}{\sqrt{5}-\sqrt{2}}$

(d) $\dfrac{\sqrt{3}-2}{\sqrt{3}+1}$ (e) $\dfrac{2\sqrt{6}+\sqrt{3}}{\sqrt{6}-\sqrt{3}}$ (f) $\dfrac{4-\sqrt{10}}{\sqrt{10}-3}$

5. Simplify the radicals first, then rationalize and simplify.

(a) $\dfrac{6}{\sqrt{12}+\sqrt{6}}$ (b) $\dfrac{\sqrt{20}-2}{\sqrt{5}+1}$ (c) $\dfrac{\sqrt{8}}{\sqrt{18}-\sqrt{12}}$

6. A student simplified $\dfrac{5}{2+\sqrt{3}}$ by multiplying the numerator and denominator by $\sqrt{3}$.

(a) Explain why this does not rationalize the denominator.

(b) Simplify $\dfrac{5}{2+\sqrt{3}}$ correctly.

7. Rationalize the denominator. State the restrictions on the variable.

(a) $\dfrac{1}{\sqrt{x}-3}$

(b) $\dfrac{4}{2+\sqrt{y}}$

(c) $\dfrac{\sqrt{a}}{\sqrt{a}+\sqrt{5}}$

(d) $\dfrac{\sqrt{m}+1}{\sqrt{m}-1}$

8. Simplify.

(a) $\dfrac{2}{\sqrt{3}-1}-\sqrt{3}$

(b) $\dfrac{1}{\sqrt{5}-2}-\dfrac{1}{\sqrt{5}+2}$

(c) $\dfrac{4}{\sqrt{2}}+\dfrac{2}{\sqrt{2}+2}$

(d) $(3+\sqrt{2})(3-\sqrt{2})\div(\sqrt{7}-2)$

9. A triangle has an area of 4 cm^2 and a base of $(\sqrt{6}+\sqrt{2})$ cm. Determine its exact height with a rational denominator.

10. A rectangle has an area of 6 m^2 and a length of $(3+\sqrt{3})$ m. Determine its exact width and show that its perimeter is a whole number.

11. Determine the exact slope of the line through $(1,2)$ and $(\sqrt{2},2+\sqrt{2})$, with a rational denominator.

12. The **silver ratio** is $1+\sqrt{2}$. Show that its reciprocal is 2 less than the silver ratio.

13. Decide whether each value is rational or irrational. Show your work.

(a) $\dfrac{1}{3-\sqrt{8}}+\dfrac{1}{3+\sqrt{8}}$

(b) $\dfrac{1}{3-\sqrt{8}}-\dfrac{1}{3+\sqrt{8}}$

14. When $\dfrac{3}{\sqrt{5}-\sqrt{2}}$ is written with a rational denominator, the result is

A. $\sqrt{5}+\sqrt{2}$ **B.** $\sqrt{5}-\sqrt{2}$ **C.** $\dfrac{3\sqrt{5}+3\sqrt{2}}{7}$ **D.** $\dfrac{\sqrt{5}+\sqrt{2}}{3}$

15. For $x \geq 0$, $y \geq 0$, $x \neq y$, the expression $\dfrac{\sqrt{x}}{\sqrt{x}-\sqrt{y}}$ with a rational denominator is

A. $\dfrac{x-\sqrt{xy}}{x-y}$ **B.** $\dfrac{x+\sqrt{xy}}{x-y}$ **C.** $\dfrac{x+\sqrt{xy}}{x+y}$ **D.** $\dfrac{\sqrt{x}+\sqrt{y}}{x-y}$

16. When $\dfrac{14}{2\sqrt{3}-\sqrt{5}}$ is rationalized and simplified, the result is

A. $4\sqrt{3}-2\sqrt{5}$ **B.** $\dfrac{14(2\sqrt{3}+\sqrt{5})}{17}$ **C.** $4\sqrt{3}+2\sqrt{5}$ **D.** $28\sqrt{3}+14\sqrt{5}$

17. Numerical Response The expression $\dfrac{12}{3-\sqrt{3}}$ can be written as $a+b\sqrt{3}$, where a and b are whole numbers. The value of ab is _____.

18. Numerical Response The value of $\dfrac{\sqrt{3}+1}{\sqrt{3}-1}+\dfrac{\sqrt{3}-1}{\sqrt{3}+1}$ is _____.

Lesson 7: Unit 2 Review

Assignment

1. Express each radical as a mixed radical in simplest form.
(a) $\sqrt{98}$ (b) $\sqrt{180}$ (c) $\sqrt[3]{-108}$ (d) $\sqrt[4]{162}$ (e) $\sqrt[3]{375}$ (f) $\frac{1}{2}\sqrt{96}$

2. State the restriction on the variable for which each expression is defined.
(a) $\sqrt{x+6}$ (b) $\sqrt{12-3x}$ (c) $\sqrt[3]{2x+1}$

(d) $4\sqrt{5x}$ (e) $\sqrt[4]{7-2x}$ (f) $\dfrac{3}{\sqrt{x+2}}$

3. Simplify.
(a) $3\sqrt{20}-\sqrt{45}+2\sqrt{80}$ (b) $\sqrt{72}-\sqrt{50}+\sqrt{27}-\sqrt{12}$

(c) $2\sqrt[3]{24}-\sqrt[3]{81}+\sqrt[3]{375}$ (d) $\sqrt{48a^3}+a\sqrt{75a}-2\sqrt{3a^3},\ \ a\geq 0$

4. Express each as an entire radical. Assume $a, b \geq 0$.
(a) $6\sqrt{3}$ (b) $-2\sqrt[3]{9}$ (c) $3\sqrt[4]{2}$

(d) $\frac{2}{3}\sqrt{18}$ (e) $2a\sqrt{5a}$ (f) $-4b^2\sqrt[3]{b}$

5. Expand and simplify. In (f), $x \geq 0$.

(a) $-2\sqrt{5}\left(3\sqrt{10}-\sqrt{15}\right)$ (b) $\left(4\sqrt{2}+\sqrt{3}\right)\left(\sqrt{2}-2\sqrt{3}\right)$ (c) $\left(\sqrt{7}+2\sqrt{5}\right)^2$

(d) $\left(2\sqrt{6}-3\right)\left(2\sqrt{6}+3\right)$ (e) $\sqrt[3]{4}\left(\sqrt[3]{6}-3\sqrt[3]{2}\right)$ (f) $\sqrt{2x}\left(\sqrt{8x}-\sqrt{6}\right)$

6. Without a calculator, arrange from greatest to least: $2\sqrt{19},\ 5\sqrt{3},\ 4\sqrt{5},\ 6\sqrt{2},\ \sqrt{78}$. Show your method.

7. Rationalize the denominator and simplify. In (g), $x > 0$.

(a) $\dfrac{10}{\sqrt{5}}$ (b) $\dfrac{9}{2\sqrt{6}}$ (c) $\dfrac{\sqrt{7}}{\sqrt{12}}$ (d) $\dfrac{12}{\sqrt[3]{3}}$

(e) $\sqrt{\dfrac{3}{20}}$ (f) $\dfrac{6-\sqrt{8}}{\sqrt{2}}$ (g) $\dfrac{5}{\sqrt{2x}}$

8. Simplify. Assume all variables are non-negative.

(a) $\sqrt{63x^6}$ (b) $\sqrt{98a^3b^4}$ (c) $\sqrt[3]{-250m^8}$ (d) $-2\sqrt{45x^5y}$

9. Rationalize the denominator and simplify. In (e), $x \geq 0$ and $x \neq 4$.

(a) $\dfrac{6}{\sqrt{5}+1}$ (b) $\dfrac{\sqrt{3}}{\sqrt{6}-\sqrt{3}}$ (c) $\dfrac{4-\sqrt{2}}{4+\sqrt{2}}$

(d) $\dfrac{2\sqrt{5}+\sqrt{3}}{\sqrt{5}-2\sqrt{3}}$ (e) $\dfrac{\sqrt{x}}{\sqrt{x}-2}$

10. Simplify. In (d), $x > 0$.

(a) $\dfrac{18\sqrt{30}}{6\sqrt{5}}$ (b) $\dfrac{\sqrt{48}+\sqrt{12}}{\sqrt{3}}$ (c) $\dfrac{4\sqrt{14}-10\sqrt{21}}{2\sqrt{7}}$ (d) $\dfrac{\sqrt{75x^5}}{\sqrt{3x}}$

11. A right triangle has legs of length $3\sqrt{2}$ cm and $\sqrt{6}$ cm.

(a) Determine the exact length of the hypotenuse in simplest form.

(b) Determine the exact perimeter.

(c) Determine the exact area.

12. A rectangle has an area of 24 m^2 and a width of $\left(\sqrt{7}-\sqrt{3}\right)$ m.

(a) Show that its length is $\left(6\sqrt{7}+6\sqrt{3}\right)$ m.

(b) Determine the exact perimeter.

13. A student made each error below. Explain the mistake and give the correct value in simplest form.

(a) $\sqrt{9+16} = \sqrt{9} + \sqrt{16} = 7$ (b) $\left(\sqrt{3}+\sqrt{2}\right)^2 = 3+2 = 5$

14. (a) Classify each number as rational or irrational. $\sqrt{\dfrac{9}{16}},\ \sqrt[3]{100},\ \sqrt{18}\cdot\sqrt{2},\ \dfrac{\sqrt{45}}{\sqrt{5}},\ 5-\sqrt{9},\ \sqrt{8}+\sqrt{2}$

(b) Give two irrational numbers whose product is rational, and two irrational numbers whose sum is rational.

15. (a) Show that $x = 2-\sqrt{3}$ is a solution of the equation $x^2 - 4x + 1 = 0$.

(b) Show that $\dfrac{1}{2-\sqrt{3}} = 2+\sqrt{3}$.

16. $3\sqrt[3]{4}$, written as an entire radical, is
A. $\sqrt[3]{12}$ **B.** $\sqrt[3]{36}$ **C.** $\sqrt[3]{108}$ **D.** $\sqrt[3]{1728}$

17. The expression $\sqrt{9-2x}$ is defined for
A. $x \geq \frac{9}{2}$ **B.** $x \leq \frac{9}{2}$ **C.** $x \geq -\frac{9}{2}$ **D.** $x \leq -\frac{9}{2}$

18. Numerical Response A rectangle has length $\sqrt{48}$ cm and width $\sqrt{27}$ cm. Its perimeter can be written as the entire radical $\sqrt{k}$ cm. The value of k is _____.

19. Which of the following is **not** equal to $4\sqrt{7}$?
A. $\sqrt{112}$ **B.** $2\sqrt{28}$ **C.** $4\sqrt{14}$ **D.** $\frac{1}{2}\sqrt{448}$

20. $5\sqrt{12} - 2\sqrt{27} + \sqrt{3}$ simplifies to
A. $3\sqrt{3}$ **B.** $4\sqrt{3}$ **C.** $5\sqrt{3}$ **D.** $5\sqrt{6}$

21. Numerical Response The expression $\sqrt{3}\left(\sqrt{24}-2\sqrt{6}\right)+\sqrt{2}\left(\sqrt{18}+\sqrt{10}\right)$ can be written in simplest form as $a+b\sqrt{c}$. The value of $a+b+c$ is _____.

22. $\left(\sqrt{5}-\sqrt{2}\right)^2$ is equal to

A. 3 **B.** 7 **C.** $7-2\sqrt{10}$ **D.** $3-2\sqrt{10}$

23. $\dfrac{12}{\sqrt{6}}$, written with a rational denominator, is

A. $\sqrt{2}$ **B.** $2\sqrt{6}$ **C.** $6\sqrt{2}$ **D.** $12\sqrt{6}$

24. Numerical Response When $5\sqrt[3]{2}$ is written as the entire radical $\sqrt[3]{k}$, the value of k is _____.

25. $\left(3\sqrt{2}+4\right)\left(3\sqrt{2}-4\right)$ is equal to

A. 2 **B.** -10 **C.** 34 **D.** $2-24\sqrt{2}$

26. Which number is irrational?

A. $\sqrt{\dfrac{49}{4}}$ **B.** $\sqrt[3]{-8}$ **C.** $\sqrt{12}\cdot\sqrt{3}$ **D.** $\sqrt{8}+\sqrt{2}$

27. Numerical Response The expression $\dfrac{14}{3-\sqrt{2}}$ can be written in simplest form as $a+b\sqrt{c}$. The value of abc is _____.